

- vii. **Trade Size Bounds:** defines the maximum and/or minimum number of shares to execute in each period through completion of the order. For example, investors may wish to trade at least 100 shares in each period and no more than say 25,000 shares.
- viii. **Residual Size Bounds:** defines the maximum and/or minimum position sizes (e.g., unexecuted shares) at different points in time. The investor may wish to give the optimizer some leeway on the solution but within a user specified tolerance band. For example, the investor may require one-half of the order to be executed within the first two hours of the trading day.

The last two constraints are often stated as *cash-balancing* constraints. However, the term cash balancing is a very vague term in the industry and has two different meanings. These cash-balancing constraints are “self-financing,” and “risk-management.”

- ix. **Self-Financing:** The self-financing constraint is used by investors who are looking to have their sell orders finance their buy orders. This constraint manages the cash transactions throughout the day. For example, if this constraint is positive it indicates that they have bought more than they have sold, and therefore will need to pay incremental dollars. If this constraint is negative it indicates that they will have sold more than they have purchased, and will have incremental dollars that they will receive. Investors will often place tolerance bands on the cash position so that they will not have to provide too much additional cash for the purchases or receive too much cash back from sells. The self-financing constraint manages cash-flow from the perspective of shares already traded. It is often intended to keep the fund from having to raise cash at the end of the day in cases where the buy dollar amount was higher than the sell dollar amount.
- x. **Risk-Management:** The risk-management constraint is used by investors to manage risk throughout the trading day. Here risk is managed by the net value of the unexecuted shares. These investors believe that as long as the value of the remaining shares to be purchased is equal to the value of shares to be sold the portfolio is hedged from market movement. This constraint, however, does not incorporate the sensitivity to the market. For example, if the investor is buying a list of high beta technology stocks and selling a list of low beta consumer staples they may not be hedged from market movement. If the market goes up the prices of the technology stocks are likely to go up more than the consumer staples stocks thus requiring the investor to provide additional cash at the end of the day and incur a higher trading cost. The risk-management constraint manages cash-flow from the perspective of